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Laboratory 2 FDSAI

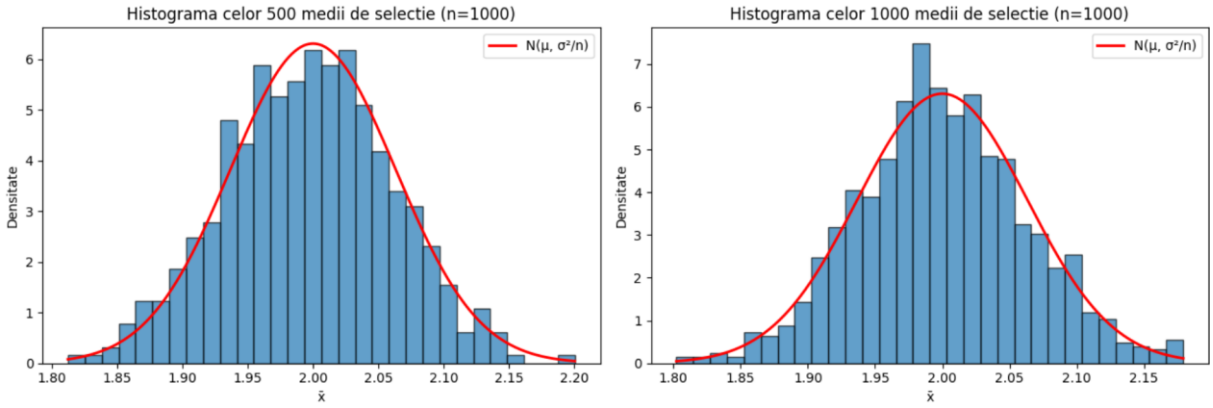
Ex 1: Distribution of the sample means

I chose the exponential distribution with mean μ = 2 and standard deviation σ = 2. The choice is deliberate. The exponential distribution is strongly right-skewed, so it is very far from a bell shape. If the sample means nevertheless end up normally distributed, the effect of the Central Limit Theorem is clearly visible.

The result is a table in which each row t = 1, …, T is a sample of 1000 values (columns X1, …, X1000) and the last column holds the mean of that sample.

```
--- T = 500 esantioane, n = 1000 ---
           X1        X2        X3     X1000     x_bar
t=1  4.808417  4.672379  4.769522  0.770389  2.028716
t=2  0.455538  0.634410  0.255694  0.850169  2.031244
t=3  2.519497  2.335460  0.305659  1.901677  2.003320
t=4  5.249464  3.002088  2.002670  2.145129  1.937229
t=5  0.240679  0.457523  0.118307  3.850721  1.837095
Media mediilor = 1.9962 (teoretic 2.0), abaterea standard = 0.0632 (teoretic 0.0632)

--- T = 1000 esantioane, n = 1000 ---
           X1        X2        X3     X1000     x_bar
t=1  2.328470  0.317695  0.330020  0.015640  2.114093
t=2  1.656754  3.909814  0.273206  2.218537  1.982805
t=3  0.592781  0.123863  0.129746  0.268224  1.982166
t=4  3.542914  2.749097  1.505579  0.889561  1.992059
t=5  3.559285  4.702826  2.988732  0.730562  2.073480
Media mediilor = 1.9995 (teoretic 2.0), abaterea standard = 0.0630 (teoretic 0.0632)
```



Numerical results

T	Mean of the means	Theoretical value (μ)	Std. dev. of the means	Theoretical value (σ/√n)
500	1.9962	2.0000	0.0632	0.0632
1000	1.9995	2.0000	0.0630	0.0632

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Interpretation

Although the original data follows an asymmetric exponential distribution, the Central Limit Theorem dictates that the sample means adopt a normal bell shape for a sufficiently large sample size (n). The empirical means (1.9962 for $T = 500$ and 1.9995 for $T = 1000$) are virtually identical to the theoretical mean ($\mu = 2$), confirming that the sample mean is an unbiased estimator whose accuracy increases in line with the Law of Large Numbers.

The standard deviation of the sample means matches the theoretical value ($\sigma/\sqrt{n}$), demonstrating a much higher concentration around the center than the individual values. Finally, increasing the number of repetitions (T) merely improves the histogram's approximation accuracy, whereas the limiting shape of the distribution is governed solely by the sample size (n).

Ex 2: Preparing the teachingratings dataset

The dataset contains student evaluations of university professors together with characteristics of the professors and of the courses. It has 463 observations (evaluated courses) and 19 variables, covering 94 distinct professors (the variable prof takes values from 1 to 94). A professor appears on several rows, one for each course taught.

```
=== a) Overview ===
Dimensiune: (463, 19)
  minority  age  gender credits    beauty  eval division native tenure  \
0      yes   36  female    more  0.289916   4.3    upper    yes    yes
1      yes   36  female    more  0.289916   3.7    upper    yes    yes
2      yes   36  female    more  0.289916   3.6    upper    yes    yes
3      yes   36  female    more  0.289916   4.4    upper    yes    yes
4       no   59    male    more -0.737732   4.5    upper    yes    yes

   students  allstudents  prof  PrimaryLast  vismin  female  single_credit  \
0        24           43     1            0       1       1              0
1        86          125     1            0       1       1              0
2        76          125     1            0       1       1              0
3        77          123     1            1       1       1              0
4        17           20     2            0       0       0              0

   upper_division  English_speaker  tenured_prof
0               1                1             1
1               1                1             1
2               1                1             1
3               1                1             1
4               1                1             1
```

```
RangeIndex: 463 entries, 0 to 462
Data columns (total 19 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   minority         463 non-null    object
 1   age              463 non-null    int64
 2   gender           463 non-null    object
 3   credits          463 non-null    object
 4   beauty           463 non-null    float64
 5   eval             463 non-null    float64
 6   division         463 non-null    object
 7   native           463 non-null    object
 8   tenure           463 non-null    object
 9   students         463 non-null    int64
 10  allstudents      463 non-null    int64
 11  prof             463 non-null    int64
 12  PrimaryLast      463 non-null    int64
 13  vismin           463 non-null    int64
 14  female           463 non-null    int64
 15  single_credit    463 non-null    int64
 16  upper_division   463 non-null    int64
 17  English_speaker  463 non-null    int64
 18  tenured_prof     463 non-null    int64
```

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A useful remark for later analysis: the last seven columns are largely redundant with the text variables, so models should not use both versions at the same time, to avoid perfect collinearity. In addition, students and allstudents have a mean far above the median (36.6 vs. 23 and 55.2 vs. 29), a sign of a strongly right-skewed distribution with a few very large courses.

b) Encoding the categorical variables

```
=== b) Encoding ===
minority: {'no': 0, 'yes': 1}
gender: {'female': 0, 'male': 1}
credits: {'single': 0, 'more': 1}
division: {'lower': 0, 'upper': 1}
native: {'no': 0, 'yes': 1}
tenure: {'no': 0, 'yes': 1}
   minority  age  gender  credits    beauty  eval  division  native  tenure  \
0         1   36       0        1  0.289916   4.3         1       1       1
1         1   36       0        1  0.289916   3.7         1       1       1
2         1   36       0        1  0.289916   3.6         1       1       1
3         1   36       0        1  0.289916   4.4         1       1       1
4         0   59       1        1 -0.737732   4.5         1       1       1

   students  allstudents  prof  PrimaryLast  vismin  female  single_credit  \
0        24           43     1            0       1       1              0
1        86          125     1            0       1       1              0
2        76          125     1            0       1       1              0
3        77          123     1            1       1       1              0
4        17           20     2            0       0       0              0

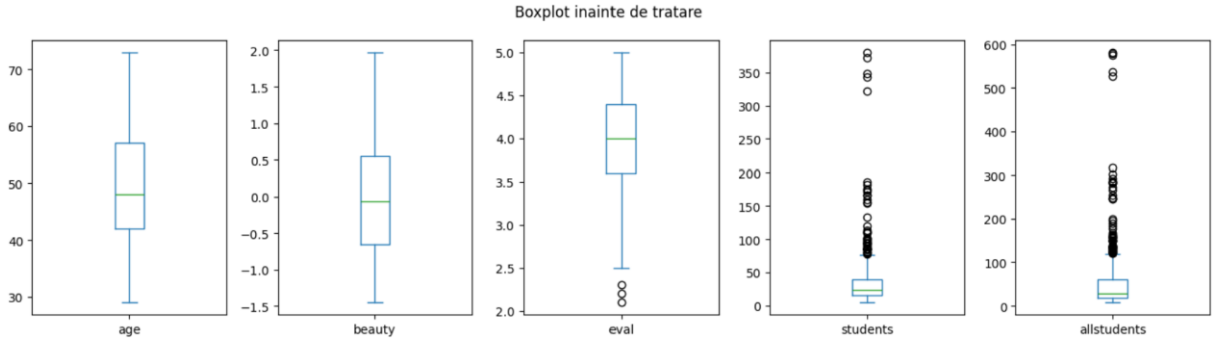
   upper_division  English_speaker  tenured_prof
0               1                1             1
1               1                1             1
2               1                1             1
3               1                1             1
4               1                1             1
```

For variables with two categories, binary encoding (equivalent to one-hot encoding with one category dropped) is sufficient and does not introduce an artificial order between categories. Checking the data types after encoding shows that all columns are now numeric (int64 or float64). The variable prof was kept as an identifier and is not used as a numeric variable.

c) Identifying and handling outliers

I preferred this method over the Z-score because students and allstudents are strongly skewed, while the Z-score assumes an approximately normal distribution and relies on the mean and standard deviation, which are themselves distorted by extreme values. The IQR rule is based on quartiles, which are robust, requires no assumption about the shape of the distribution, and is easy to interpret and to check on a boxplot.

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Interpretation

Age and beauty have no outliers. eval has only three, all at the lower end (about 2.1–2.3), visible on the boxplot below the lower whisker. These are real and plausible ratings on a scale from 1 to 5, not measurement errors, so they were kept, as were the beauty values. The main issue is with students and allstudents: many courses exceed the upper limit, and the maxima (380 and 581) are several times larger than the fence. They most likely represent introductory courses with hundreds of students, so they are real but atypical values.

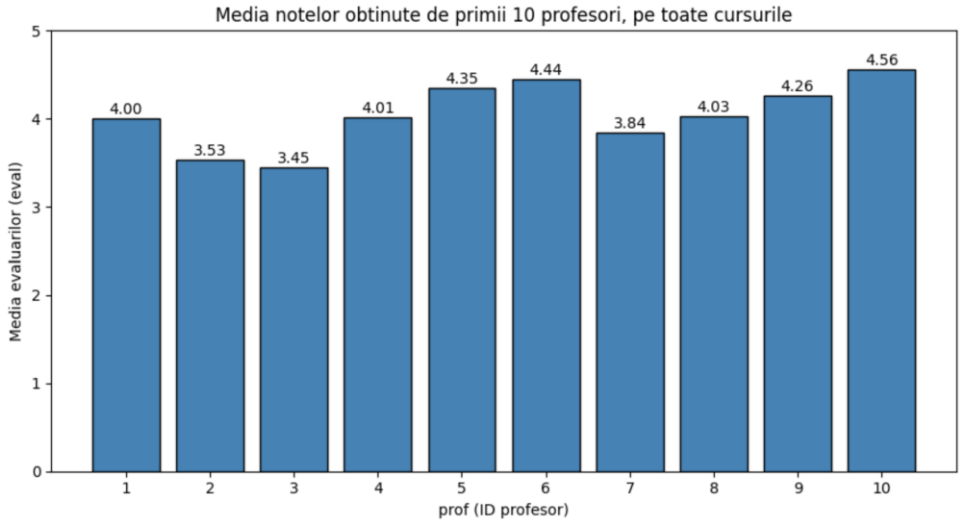
Ex 3: Average rating per professor

Since a professor appears on several rows (one per course), the average rating is obtained by grouping the rows by prof and computing the mean of the eval column. The first 10 distinct professors (1–10) were used, giving the averages in the table below. The variable eval was not modified in the previous step, so the averages use the original ratings.

```
Dupa winsorizare:
             count       mean        std   min   25%   50%   75%    max
age          463.0  48.365011   9.802742  29.0  42.0  48.0  57.0   73.0
students     463.0  30.617711  21.427459   5.0  15.0  23.0  40.0   77.5
allstudents  463.0  44.254860  34.928852   8.0  19.0  29.0  60.0  121.5

Media evaluarilor pentru primii 10 profesori:
prof
1     4.000000
2     3.533333
3     3.450000
4     4.012500
5     4.350000
6     4.442857
7     3.840000
8     4.028571
9     4.257143
10    4.560000
```

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Interpretation

The averages of the 10 professors range from **3.45** (professor 3) to **4.56** (professor 10), so the difference between the lowest- and highest-rated professor in this group is about 1.1 points on the 1–5 scale. Relative to the overall mean of the dataset (3.998, i.e. about 4.00), six professors are above average (4, 5, 6, 8, 9 and 10), professor 1 is exactly at the average, and three are below it (2, 3 and 7). The lowest averages (professors 2 and 3, with 3.53 and 3.45) are also the only ones below 3.8.

Two caveats apply. First, the averages are computed over a different number of courses for each professor, so a professor with a single evaluated course has a less stable average than one with many courses. Second, prof is just an identifier, so the order on the x-axis has no meaning: the fact that professor 10 has the highest average does not mean that ratings increase with the professor number. The plot is descriptive and does not allow conclusions about the causes of the differences, which would require the explanatory variables in the dataset (for example beauty, age, division).